

MICHAEL RABA PORTFOLIO FALL 25

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Insertion Loss for Multichamber vs Competing Mufflers

- Goal: maximize Insertion Loss
- Insertion loss is the reduction in signal power (e.g., sound or RF) caused by inserting a device into a system, whereas transmission loss measures the total reduction in signal as it passes through a medium or barrier, independent of any added device.
- IL isolates the muffler’s effect, letting evaluate designs without being affected by other parts of the exhaust/environment

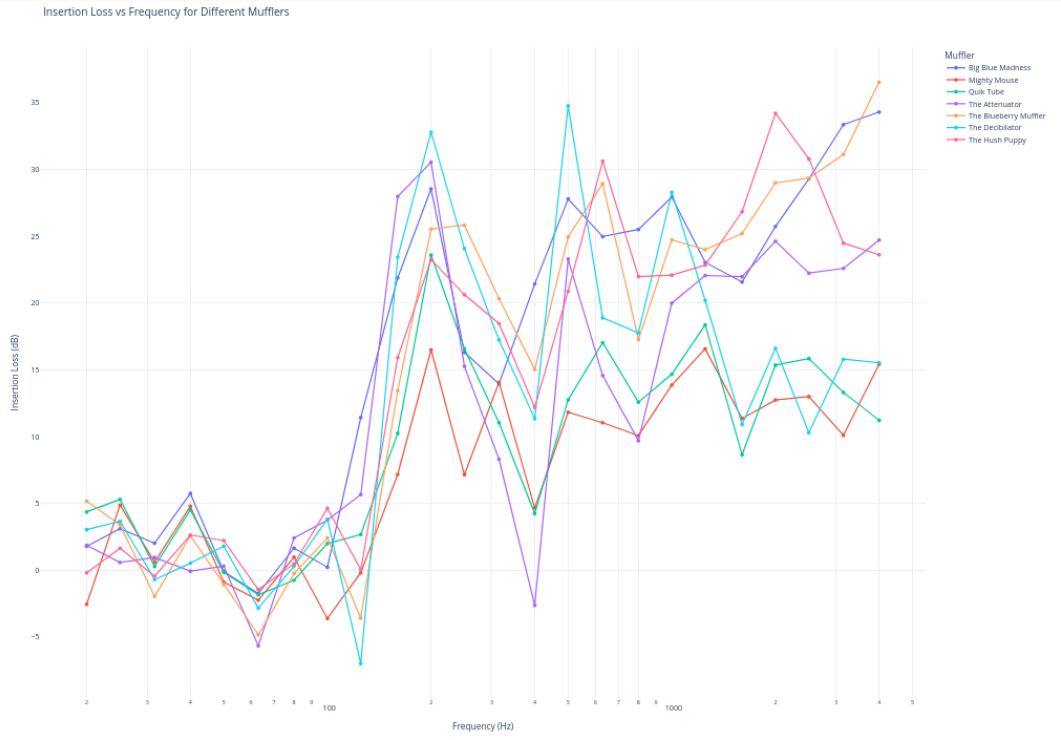


Figure 1: Insertion Loss for Muffler Competition

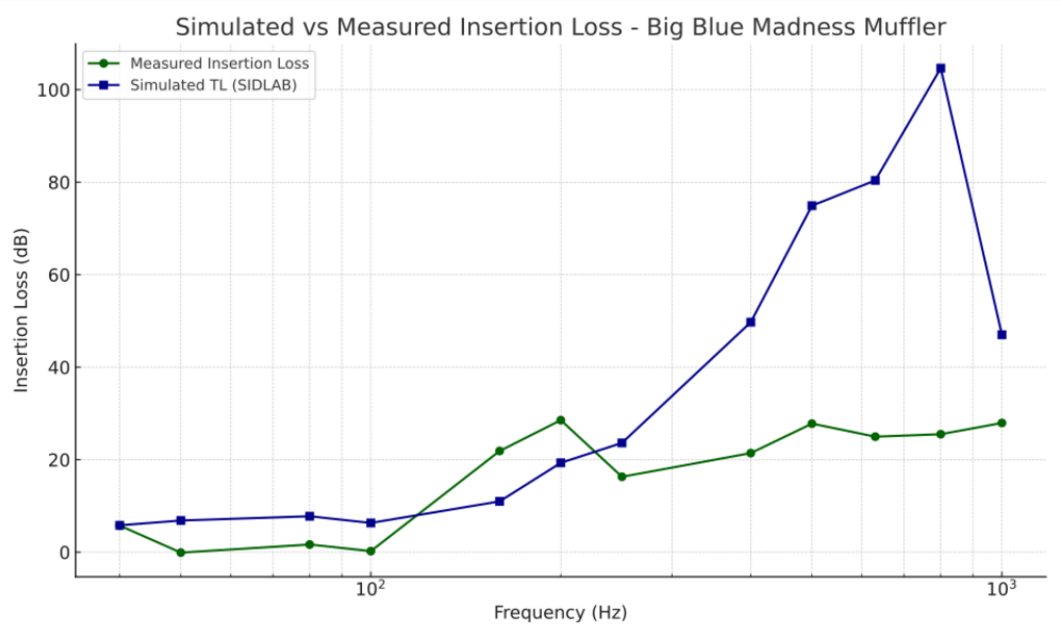


Figure 2: Sidlab 1d Insertion Loss Approximation (Transfer Matrix system)

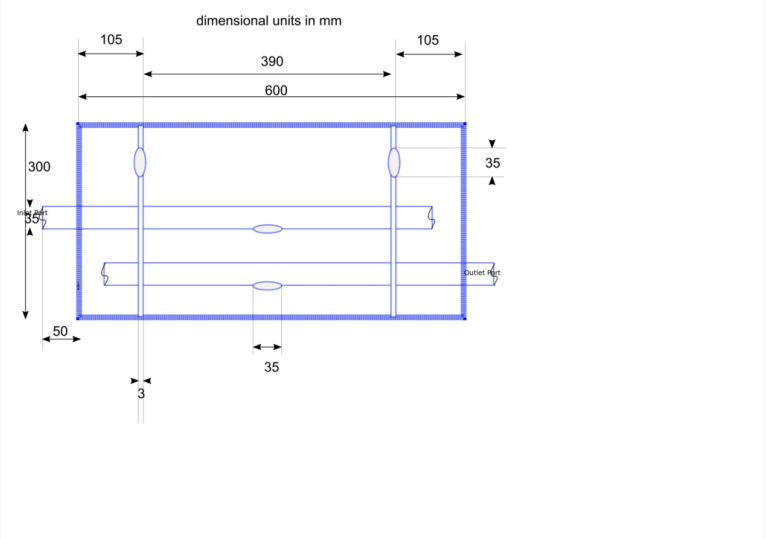


Figure 3: Schematic

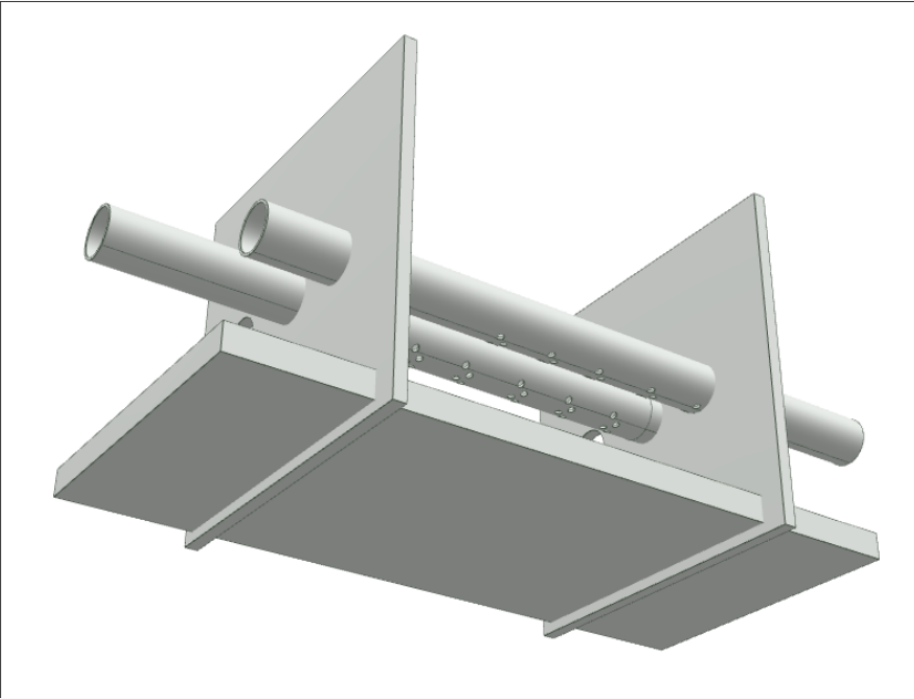


Figure 4: Multicomponent Muffler Internal Geometry (Ansys Workbench/Acoustics Module)

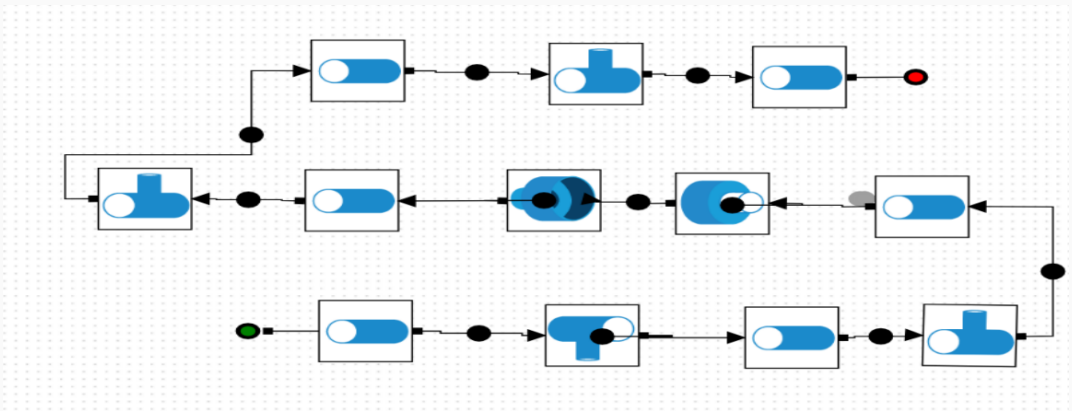


Figure 5: Sidlab 1d approximation (Transfer Matrix system)

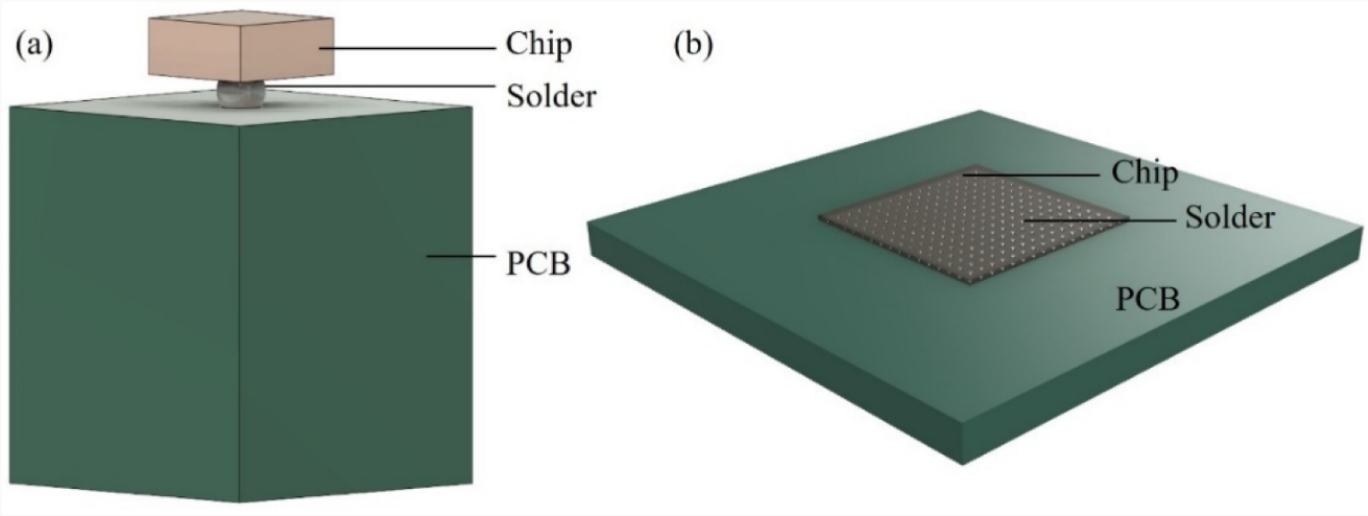


Figure 6: Anand Viscoplasticity model applied to electronics packaging (pcb).

Models Viscoplastic (Metals under Temperature/thermal cycling)

- **Broad Application:** Model metals under thermal cycling! Temperature and Strain dependance!
- Applies Anand’s unified viscoplastic framework to model solder behavior
- Anand’s model can be reduced and fitted from experiments
- transition the theory into engineering-scale implementation
- Targets solder joints in microelectronic packages (chip on PCB, soldered connections)

Viscoplastic System of Equations (Anand)

Flow Rule (Plastic Strain Rate): $\dot{\epsilon}^p = A \exp\left(-\frac{Q}{RT}\right) \left[\sinh\left(\frac{j\sigma}{s}\right)\right]^{1/m},$ (2)

Plastic strain rate increases with stress and temperature,
No explicit yield surface; flow occurs at all nonzero stresses.

Deformation Resistance Saturation s^* : $s^* = \hat{s} \left(\frac{\dot{\epsilon}^p}{A} \exp\left(\frac{Q}{RT}\right)\right)^n,$ (3)

Defines the steady-state value that s evolves toward,
Depends on strain rate and temperature.

Evolution of Deformation Resistance s : $\dot{s} = h_0 \left|1 - \frac{s}{s^*}\right|^a \text{sign}\left(1 - \frac{s}{s^*}\right) \dot{\epsilon}^p,$ (4)

Describes dynamic hardening and softening of the material,
 s evolves depending on proximity to s^* and flow activity.

{Note: Constants $A, Q, m, j, h_0, \hat{s}, n, a$ are material-specific and fitted to experimental creep/strain rate data.}

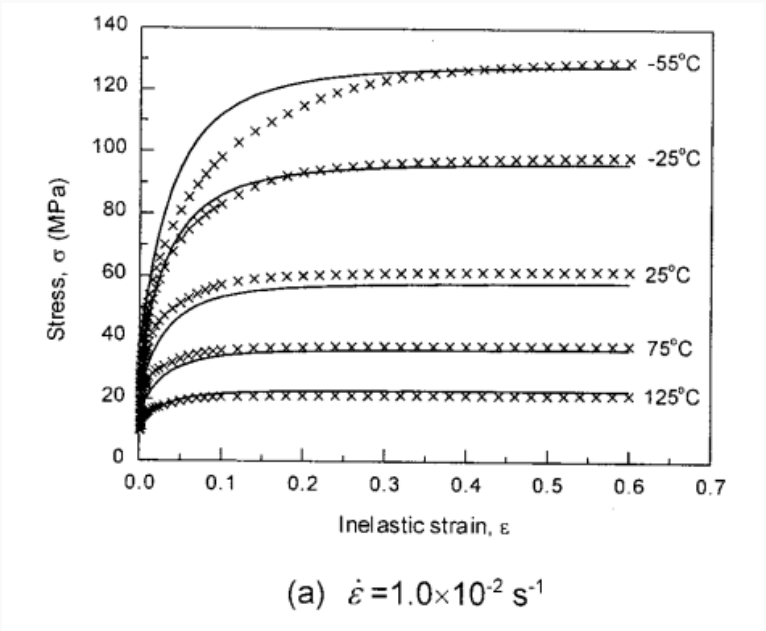


Figure 7: strain-rate and temperature dependence of solder materials (Source: Wang 2001 Paper)

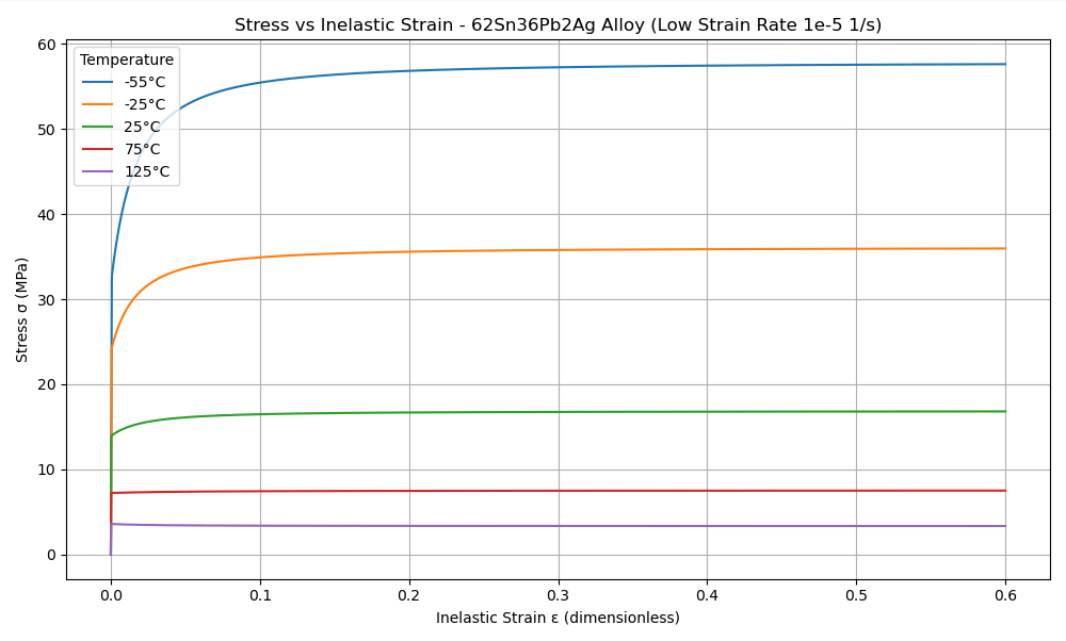


Figure 8: Running model using Forward Euler code

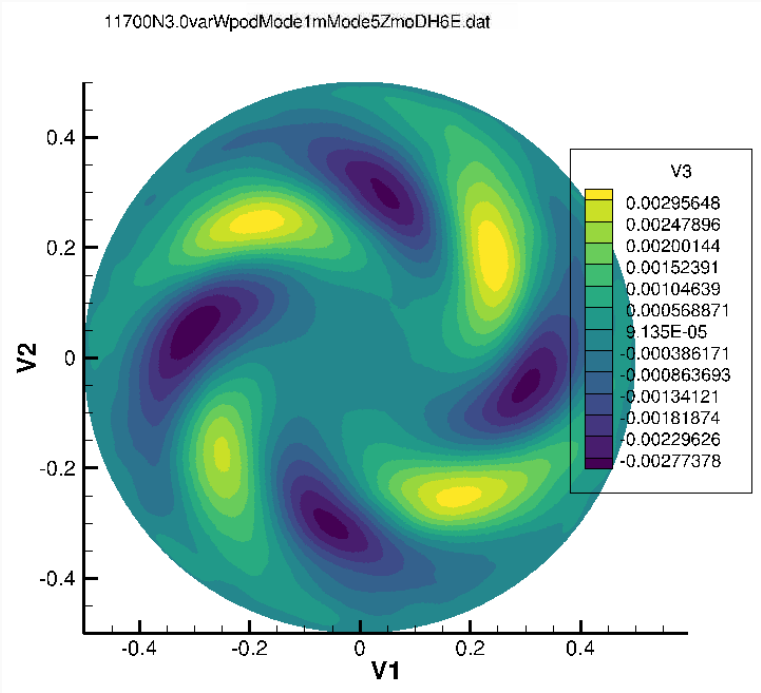
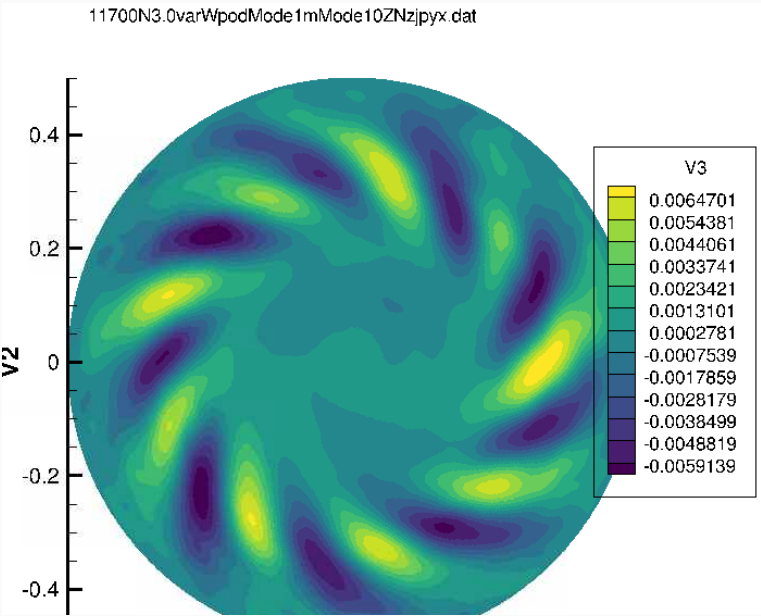


Figure 9: Decompose turbulent pipe flow (Reynolds number $Re = 5300$ and $Re = 11,700$) to Modal FFT/POD decomposition



- Idea: Reduce turbulent flow to Fft-POD so it can be easily studied. The lowest POD modes have the most energy $\therefore$ most important
- Relaminarization occurs by suppression of near-wall vortices and streak breakdown.
- Applications: rotating heat exchangers, turbomachinery, compact fluid systems.

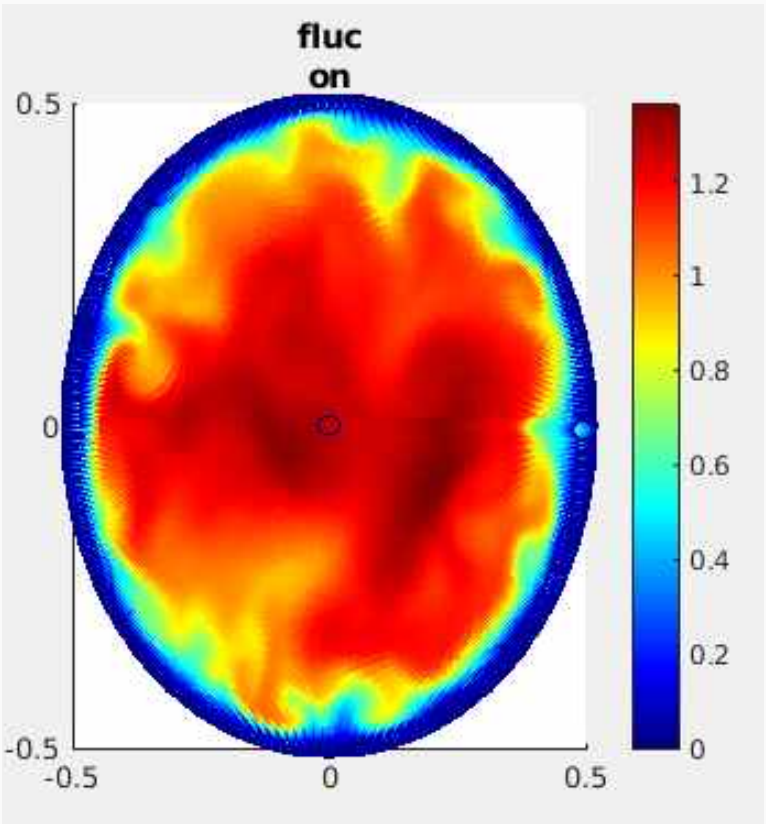


Figure 10: Fluid field in streamwise direction u_w

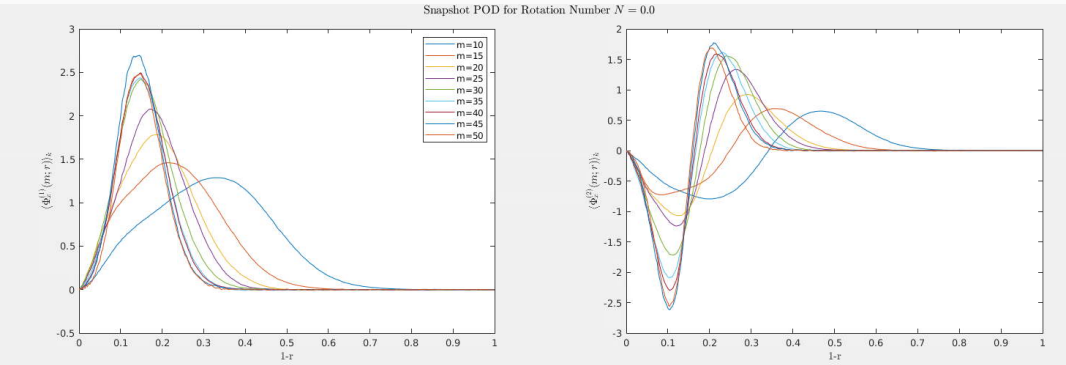


Figure 11: Turbulent Flow

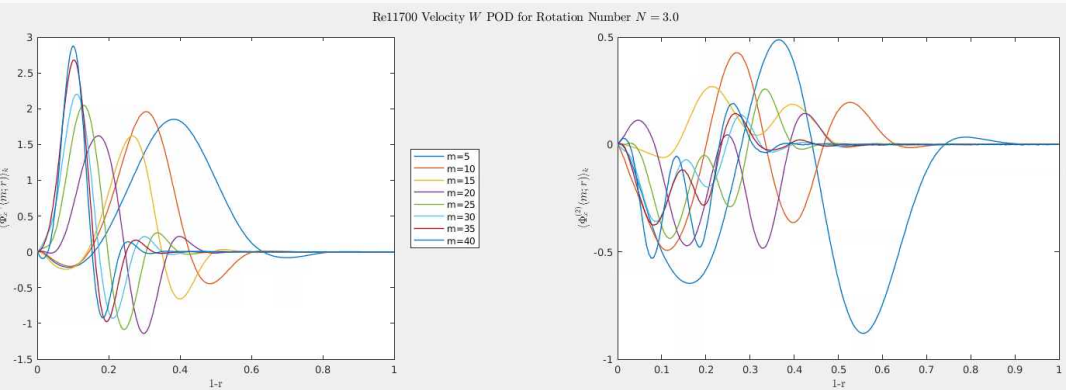


Figure 12: Relaminarized flow (Broadband Energy Distribution)

Microgripper Design

- ME 514: A microgripper is a small-scale mechanical device designed to grasp, manipulate, or position micro-sized objects, often used in biomedical, microassembly, and MEMS applications.
- Our design goal is to optimize the flexible gripper geometry to maximize Geometric Advantage (GA) while maintaining structural reliability (stress ≤ 15 MPa).
- Geometric Advantage (GA) is a performance metric defined as the ratio of the output displacement to the input displacement, indicating how effectively the gripper amplifies motion.

Set Objective & Constraints

- Objective: Maximize deformation (improve GA).
- Constraint: Stress ≤ 15 MPa.
- Optimization Method: Adaptive Single-Objective (gradient-based) to Maximize GA (via X-axis deformation)

Output Parameters

Parameter	Description
X-axis Directional Deformation	Objective (linked to GA)
Minimum Combined Stress	Constraint check
Maximum Combined Stress	Constraint check

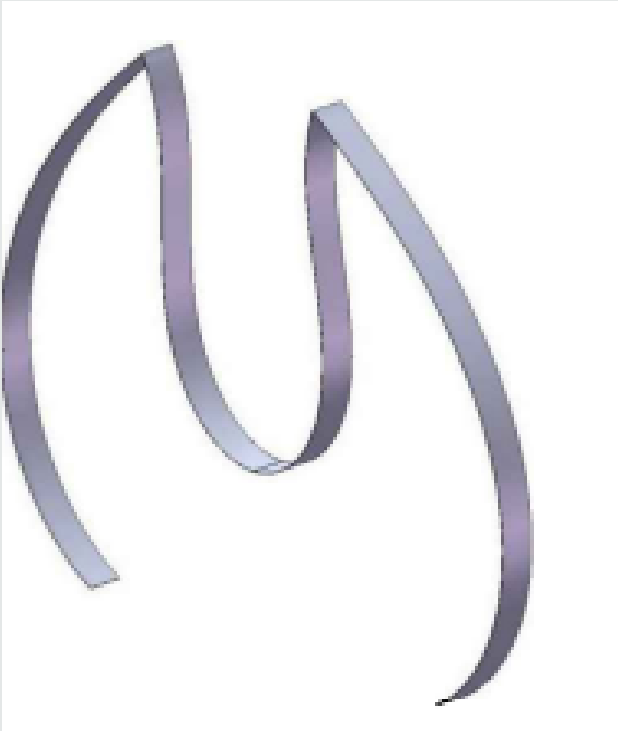


Figure 13: Microgripper shape

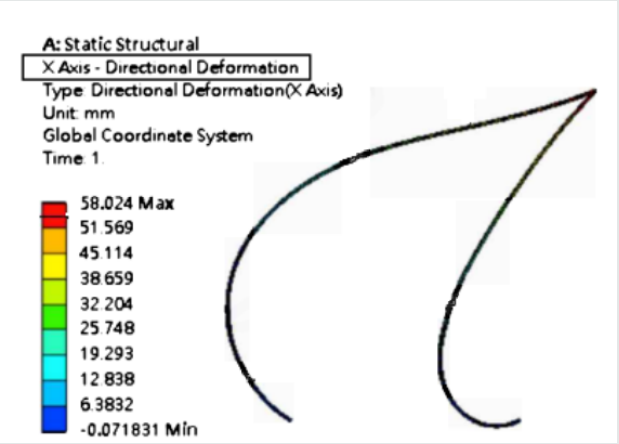


Figure 14: Input parameters for optimization

Thread and Bolt Design with Ansys

- Assume coefficeint of friction between bolt and nut = 0.2, mirror symmetry, and 2d simulation to reduce computation

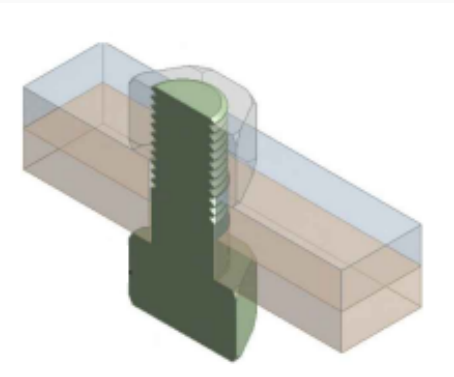


Figure 15: Threaded nut and bolt

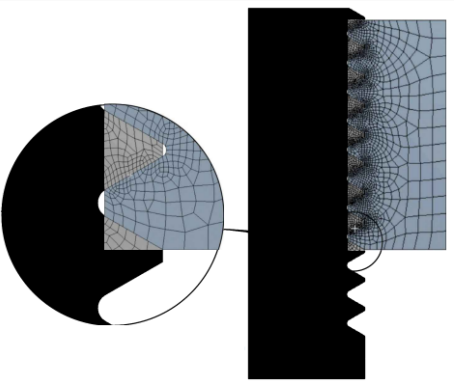


Figure 16: Mesh

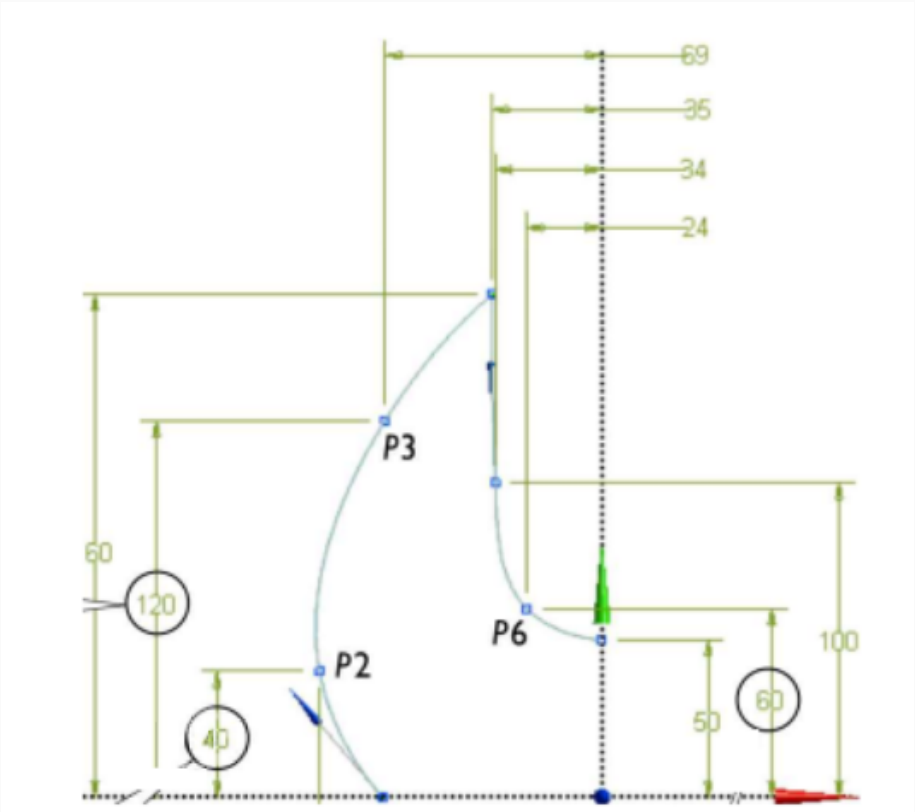
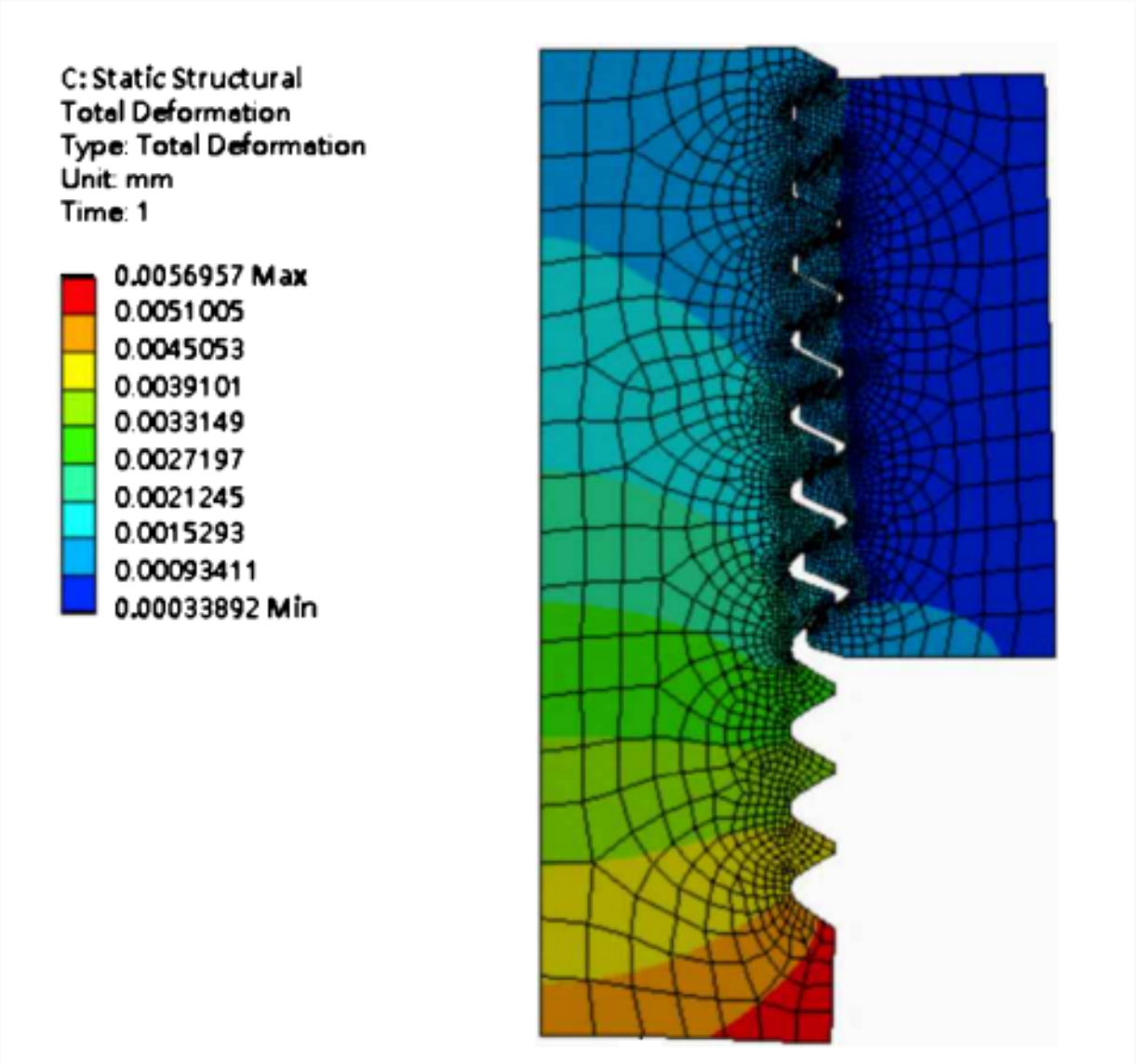


Figure 17: Design points P3, P2, P6.

Thread and Bolt Design with Ansys

Improving efficiency involves redistributing some of the load to upper threads to reduce maximum stress and enhance overall performance



most stress concentrated on a few lower threads.

uneven stress

Gripper (continued)

Goal is to optimize geometry so it could amplify motion efficiently (maximize Geometric Advantage) while keeping stresses within safe limits.

- input motion at the actuator is small, amplifying motion achieves the displacement
 - If $GA > 1$
- P2: Controls the curvature near the base of the gripper arm.
 - influences stiffness and how the arm starts to bend under load
- P3: located mid-span of gripper arm
 - largest range of motion, affects overall flexibility and amplification
 - near tip and fine tunes end geometry, impacts contact alignment and stress distribution at the gripping point